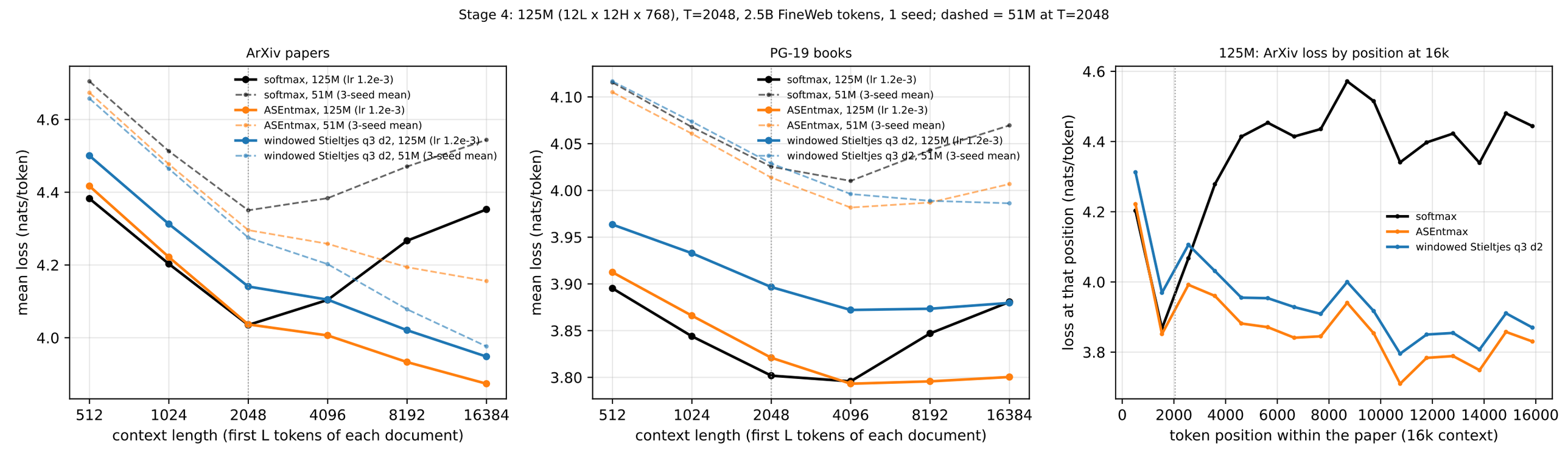
Stage 4: 125M (12L x 12H x 768), T=2048, 2.5B FineWeb tokens, 1 seed; dashed = 51M at T=2048
ArXiv papers
PG-19 books
125M: ArXiv loss by position at 16k
mean loss (nats/token)
loss at that position (nats/token)
context length (first L tokens of each document)
token position within the paper (16k context)
softmax, 125M (lr 1.2e-3)
softmax, 51M (3-seed mean)
ASEntmax, 125M (lr 1.2e-3)
ASEntmax, 51M (3-seed mean)
windowed Stieltjes q3 d2, 125M (lr 1.2e-3)
windowed Stieltjes q3 d2, 51M (3-seed mean)
softmax
ASEntmax
windowed Stieltjes q3 d2
512
1024
2048
4096
8192
16384
0
2000
4000
6000
8000
10000
12000
14000
16000
3.80
3.85
3.90
3.95
4.00
4.05
4.10
4.2
4.4
4.6